Intermittent fasting and metabolic health

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Abstract

Intermittent fasting (IF) improved glucose control in most cohorts [1], though randomized evidence remains mixed [2]. This report reviews fasting windows and their effects on insulin sensitivity and body-composition markers.

Introduction

Time-restricted eating (TRE) concentrates intake into 6-10 hour windows. Its metabolic effects depend on **energy balance**, *circadian alignment*, and adherence [1].

Key findings

Glucose and insulin

- Fasting windows reduced fasting glucose in several cohorts [1].
- Effects on HOMA-IR were smaller than expected in randomized trials [2].
- Early TRE (eating before 3 pm) showed larger improvements than late TRE [3].

Body composition

1. Weight loss was comparable to continuous energy restriction [1].
2. Fat-free mass was slightly lower in some TRE arms [2].

Conclusion

TRE is a viable strategy for glucose control; long-term randomized evidence is still limited [1][2][3].

References

[1] Intermittent fasting strategies and their effects on body weight and other cardiometabolic markers. PubMed, 2025.

[2] Effects of intermittent fasting on body composition and clinical health markers. PubMed, 2015.

[3] Effect of Intermittent Fasting on Reproductive Hormone Levels. PubMed, 2022.